Theory of Computation

Non Deterministic Finite Automata

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- **Nondeterministic Finite Accepter (NFA):**

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q: Finite set of states

Σ: Input alphabet

δ: Transition Function: $\delta : Q \times \Sigma \rightarrow 2^Q$

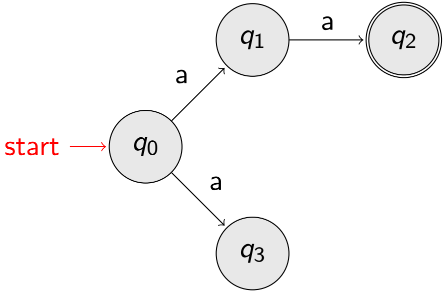
q_0: Initial State

F: Finite set of final/accepting states: $F \subseteq Q$

Nondeterministic Finite Accepter (NFA)

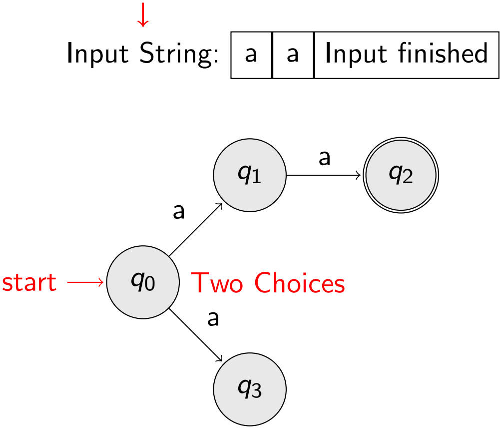
- Alphabet, $\Sigma = \{a\}$

Input String: | a | a | Input finished |



Nondeterministic Finite Accepter (NFA)

- Alphabet, $\Sigma = \{a\}$



Nondeterministic Finite Accepter (NFA)

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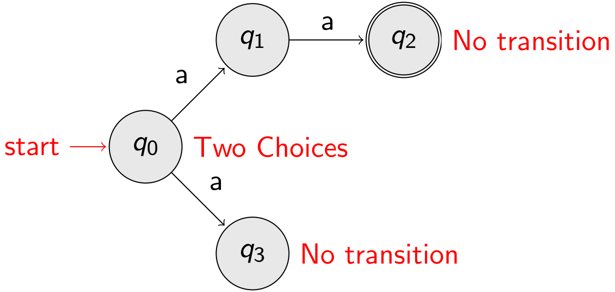
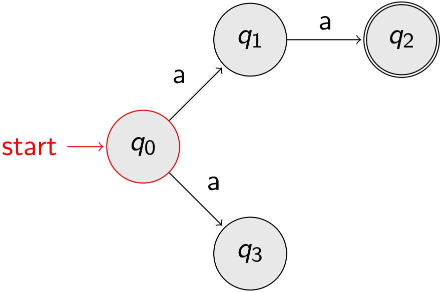
Input String: | a | a | Input finished |



- **First Choice:**

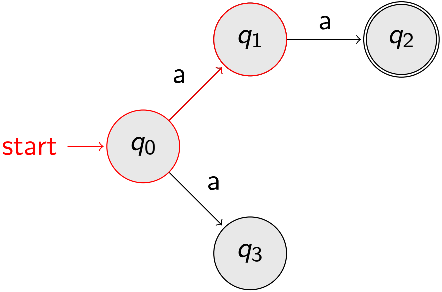
Input String: | a | a | Input finished |



Nondeterministic Finite Accepter (NFA)

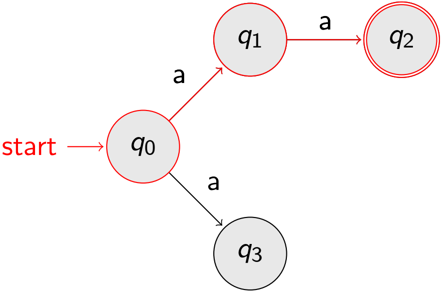
- **First Choice:**

Input String: | a | a | Input finished |



- **First Choice:**

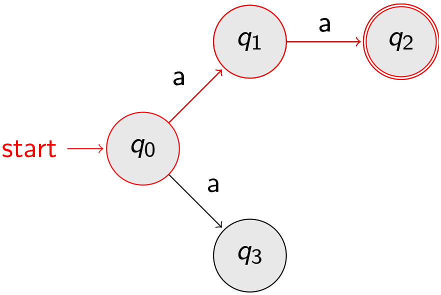
Input String:	a	a	Input finished



Nondeterministic Finite Accepter (NFA)

- **First Choice:**

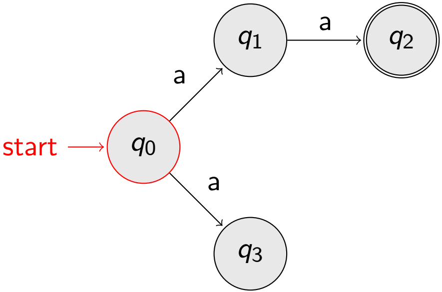
Input String: | a | a | Input finished |



All input is consumed
Accepted

- **Second Choice:**

Input String:	a	a	Input finished

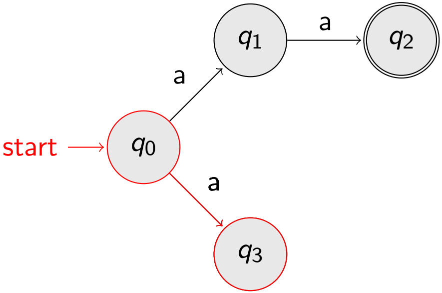


Nondeterministic Finite Accepter (NFA)

- **Second Choice:**

Input String:

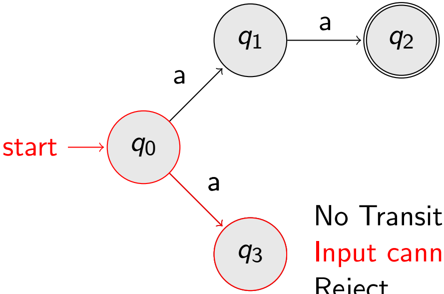
a	a	Input finished



Nondeterministic Finite Accepter (NFA)

- **Second Choice:**

Input String: | a | a | Input finished |



No Transition

Input cannot be consumed

Reject

Nondeterministic Finite Accepter (NFA)

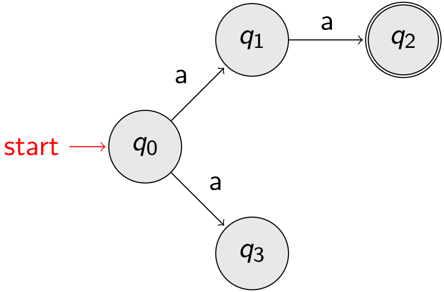
- **An NFA accepts a string:**
  - when there is a computation of the NFA that accepts the string,
  - All the input is consumed and the automaton is in a final state.

Nondeterministic Finite Accepter (NFA)

- **An NFA rejects a string:**
  - when there is no computation of the NFA that accepts the string
  - All the input is consumed and the automaton is in a non final state,
  - The input cannot be consumed
- **Language accepted:**

  Example: $L = \{aa\}$

Input String:	a	a	Input finished



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Languages Accepted by NFAs

- The language accepted by a NFA, $M = (Q, \Sigma, \delta, q_0, F)$ can be defined as follows:
  - $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \bigcap F \neq \Phi\}$
    Here,
    Q: Finite set of states
    Σ: Input alphabet
    δ: Transition function: $\delta^* : Q \times \Sigma^* \rightarrow 2^Q$
    q_0: Initial State
    F: Finite set of final states
- L(M) is a set of strings in w in Σ^* such that $\delta^*(q_0, w)$ contains atlest one accepting state.

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Languages Accepted by NFAs

- **Observations:**
  - Language accepted by a NFA, $M = (Q, \Sigma, \delta, q_0, F)$:
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Languages Accepted by NFAs

- **Observations:**
  - Language accepted by a NFA, $M = (Q, \Sigma, \delta, q_0, F)$:
    $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \bigcap F \neq \Phi\}$
  - Language rejected by a NFA, $M = (Q, \Sigma, \delta, q_0, F)$:
    $\overline{L(M)} = \{w \in \Sigma^* : \delta^*(q_0, w) \bigcap F = \Phi\}$

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Equivalence of DFA and NFA

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Formalities

- **Nondeterministic Finite Accepter with ϵ-transitions (ϵ-NFA):**

$$M = (Q, \Sigma \bigcup \{\epsilon\}, \delta, q_0, F)$$

Q: Finite set of states

$\Sigma \bigcup \{\epsilon\}$: Input alphabet, $\{\epsilon\}$ is a member of this alphabet.

δ: Transition Function: $\delta : Q \times \Sigma \bigcup \{\epsilon\} \rightarrow 2^Q$

q_0: Initial State

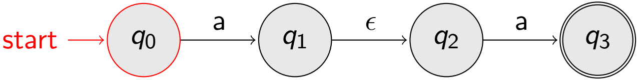
F: Finite set of final/accepting states: $F \subseteq Q$

ϵ Transitions

- ϵ **Transitions**

Input String:

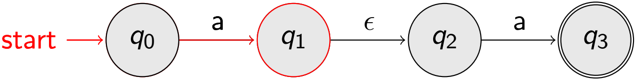
a	a	Input finished



ϵ Transitions

- **ϵ Transitions**

Input String: | a | a | Input finished |

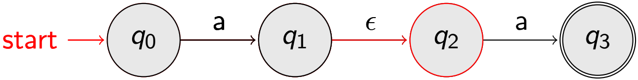


ϵ Transitions

- ϵ **Transitions**

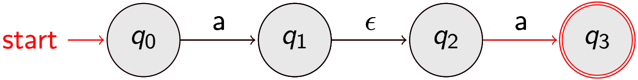
Input String: | a | a | Input finished |

read head doesn't move



- **ϵ Transitions**

Input String:	a	a	Input finished



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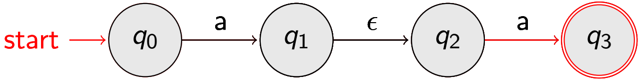
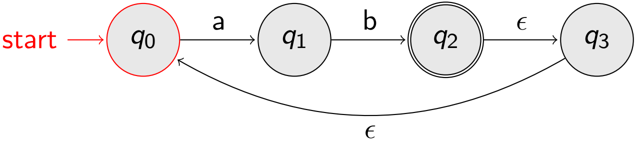
- ϵ **Transitions**

Input String:	a	a	Input finished



- **Another NFA Example**

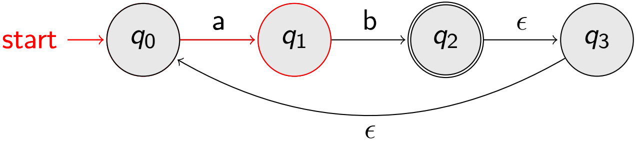
Input String:	a	b	Input finished



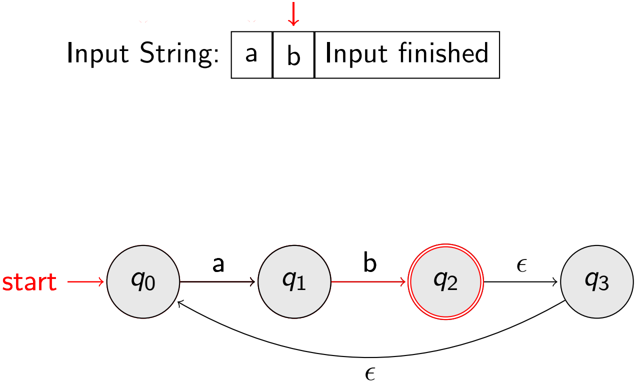
ϵ Transitions

- **Another NFA Example**

Input String:	a	b	Input finished



- **Another NFA Example**



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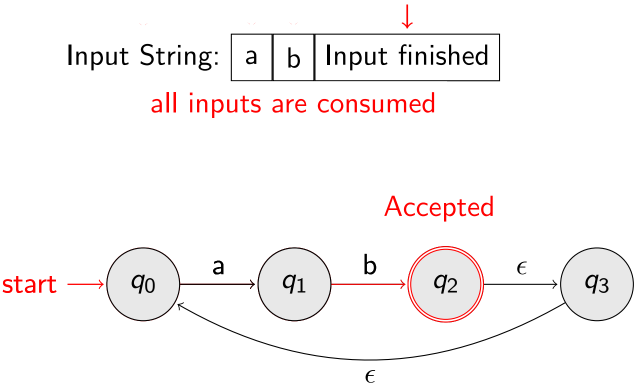
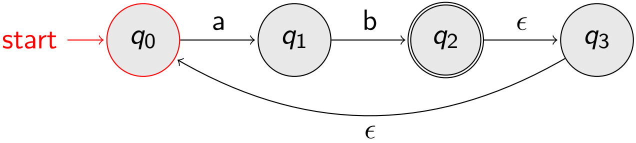
- **Another NFA Example**



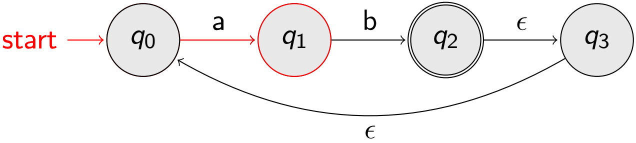
- **Another NFA Example**

Input String:	a	b	a	b	Input finished



- **Another NFA Example**

Input String: | a | b | a | b | Input finished |

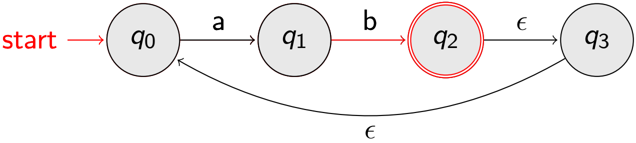


ϵ Transitions

- **Another NFA Example**

Input String:

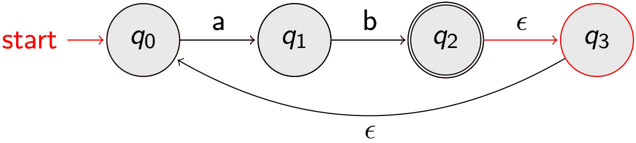
a	b	a	b	Input finished



ϵ Transitions

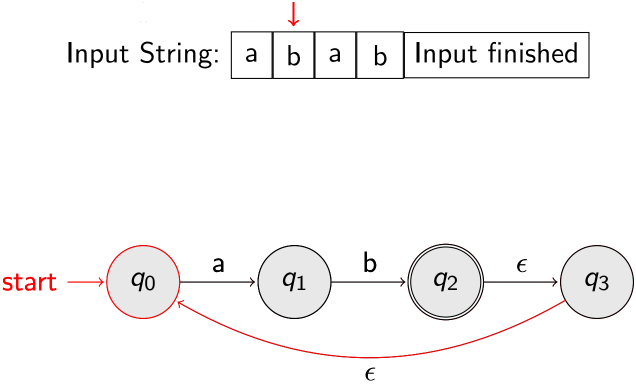
- **Another NFA Example**

Input String: | a | b | a | b | Input finished |



ϵ Transitions

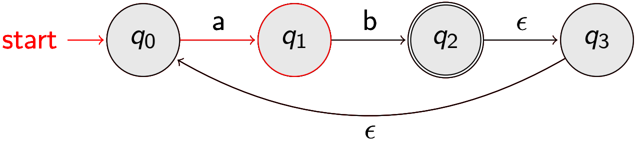
- **Another NFA Example**



ϵ Transitions

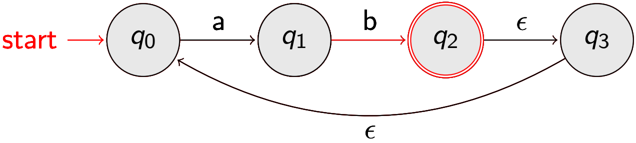
- **Another NFA Example**

Input String: | a | b | a | b | Input finished |



- **Another NFA Example**

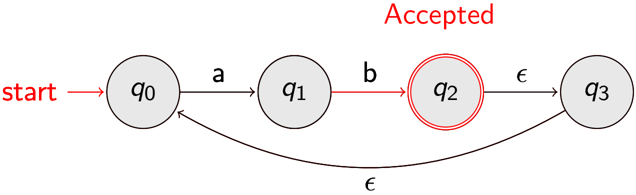
Input String:	a	b	a	b	Input finished



- **Another NFA Example**

Input String: | a | b | a | b | Input finished |

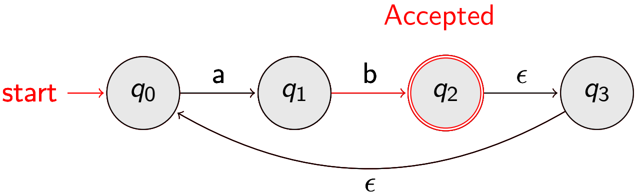
all inputs are consumed



- **Another NFA Example**

Input String:	a	b	a	b	Input finished

all inputs are consumed



- **Language accepted**

  $L = \{ab, abab, ababab, \ldots\} = \{ab\}^+$

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- Equivalence of ϵ-NFA to DFA

Equivalence of ϵ-NFA to DFA or NFA

- Equivalence of ϵ-NFA to DFA
- Equivalence of ϵ-NFA to NFA

References

- Denial I. A. Cohen Introduction to Computer Theory, Second Edition, John Wiley & Sons, 1997.
- J. E. Hopcroft, R. Motwani, & J. D. Ullman Introduction to Automata Theory, Languages, and Computation, Third Edition, Pearson, 2008.

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